

AP CALCULUS AB – UNIT 2

Differentiation: Definition and Fundamental Properties

Expanded AP-Style Review: Full Coverage and Challenge



Name: _____ Class: _____ Date: _____

Directions

This review deliberately samples every website lesson and includes a larger proportion of difficult and challenge reasoning. Part A is non-calculator. Part B is calculator-permitted. Free-response work must show setup, exact reasoning, units, and theorem conditions.

Section I – Part A: No Calculator

1. A quantity $M(t)$ is measured in kilograms and t in hours. Which is a complete interpretation of $M'(5) = -2.4$?

- (A) During the first 5 hours, mass decreases by 2.4 kilograms
 (B) At $t = 5$, time decreases at 2.4 hours per kilogram
 (C) At $t = 5$ hours, the mass is decreasing at 2.4 kilograms per hour
 (D) At $t = 5$, the mass is -2.4 kilograms

2. The limit $\lim_{x \rightarrow 3} \frac{g(x) - g(3)}{x - 3} = 7$ directly implies which statement?

- (A) g is linear
 (B) $g'(3) = 7$
 (C) $g(3) = 7$
 (D) $g'(7) = 3$

3. Which statement is always true?

- (A) If f has a vertical tangent at c , then $f'(c)$ is finite
 (B) If f is differentiable at c , then f is continuous at c
 (C) If f is continuous at c , then f is differentiable at c
 (D) If $f'(c) = 0$, then f has a local extremum at c

4. If $F(x) = x^3 \ln x$ for $x > 0$, then $F'(x)$ is

- (A) $3x^2/x$
 (B) x^3/x
 (C) $3x^2 \ln x + 1/x$
 (D) $3x^2 \ln x + x^2$

5. At $x = 1$, $f = 2$, $f' = -3$, $g = 4$, and $g' = 5$. What is $\left(\frac{f}{g}\right)'(1)$?

- (A) $-\frac{11}{8}$
 (B) $\frac{1}{2}$
 (C) $-\frac{1}{4}$
 (D) $\frac{11}{8}$

6. For $f(x) = x^3$, which expression is the average rate of change on $[a, a + h]$?

- (A) $3a^2$
 (B) $a^3 + h^3$
 (C) $3(a + h)^2$
 (D) $3a^2 + 3ah + h^2$

7. If $f(x) = \sqrt{x}$, then $f'(4)$ equals

- (A) 2
 (B) 4
 (C) $\frac{1}{4}$
 (D) $\frac{1}{2}$

8. For $f(x) = |x - 2|$, the left- and right-hand derivatives at 2 are

- (A) -1 and -1 (B) 0 and 0
(C) -1 and 1 (D) 1 and 1

9. Differentiate $f(x) = 5x^7 - 4x^2 - 4$.

- (A) $x(35x^5 - 8)$ (B) $x(5x^5 - 8)$
(C) $35x^6 - 8x + 1$ (D) $x(35x^6 - 8)$

10. Differentiate $(x^2 + 1)(5x - 1)$.

- (A) $5x^2 + 5$ (B) $10x$
(C) $2x(5x - 1)$ (D) $15x^2 - 2x + 5$

11. For $f(x) = 4x^2 - 3x$, find the average rate of change on $[3, 6]$.

- (A) 33 (B) 34
(C) -33 (D) 32

12. If $f(x) = 2x^2 - 1x + 1$, evaluate $\lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$.

- (A) 2 (B) 4
(C) 3 (D) -3

13. A table gives $f(\frac{9}{10}) = \frac{28}{5}$ and $f(\frac{11}{10}) = \frac{32}{5}$. Estimate $f'(1)$ using a symmetric difference.

- (A) 8 (B) 4
(C) $\frac{4}{5}$ (D) 2.0

14. Find $\frac{d}{dx}[\log(x)]$.

- (A) $\log(x) + 1$ (B) $-\frac{1}{x}$
(C) $\frac{1}{x}$ (D) x

15. Differentiate $\frac{x^2 + 1}{x + 2}$.

- (A) $2x$ (B) $\frac{2x}{x+2}$
(C) $\frac{-x^2 + 2x(x+2) - 1}{(x+2)^2}$ (D) $\frac{x^2 + 2x(x+2) + 1}{(x+2)^2}$

16. For $f(x) = 3x^2 + 3x$, find the average rate of change on $[1, 3]$.

- (A) 14 (B) 15
(C) -15 (D) 16

17. Find the tangent line to $y = 3x^2 - 2$ at $x = 2$.

- (A) $y = 12x - 13$ (B) $y = 13x - 16$
(C) $y = 12x - 14$ (D) $y = 12x + 34$

18. A table gives $f(\frac{9}{10}) = \frac{47}{10}$ and $f(\frac{11}{10}) = \frac{53}{10}$. Estimate $f'(1)$ using a symmetric difference.

- (A) 3 (B) $\frac{3}{5}$
(C) 1.5 (D) 6

19. Differentiate $f(x) = 3x^3 + 5x^2 - 3$.

- (A) $x(3x + 10)$ (B) $x(9x + 10)$
(C) $x(9x^2 + 10)$ (D) $9x^2 + 10x + 1$

20. At $x = 4$, $f = 4$, $g = 4$, $f' = 2$, and $g' = -1$. Find $(fg)'(4)$.

- (A) 3 (B) 4
(C) -4 (D) 5

21. For $f(x) = 2x^2 + x$, find the average rate of change on $[1, 3]$.

- (A) 9 (B) 8
(C) -9 (D) 10

22. If $f(x) = 3x^2 + 0x + 1$, evaluate $\lim_{h \rightarrow 0} \frac{f(-2 + h) - f(-2)}{h}$.

- (A) -11 (B) -12
(C) 12 (D) -13

23. A table gives $f(\frac{49}{10}) = \frac{167}{10}$ and $f(\frac{51}{10}) = \frac{173}{10}$. Estimate $f'(5)$ using a symmetric difference.

- (A) 3 (B) $\frac{3}{5}$
(C) 6 (D) 1.5

24. Find $\frac{d}{dx}[\log(x)]$.

- (A) x (B) $-\frac{1}{x}$
(C) $\frac{1}{x}$ (D) $\log(x) + 1$

Section I – Part B: Calculator Permitted

- 25.** A population is $P(t) = 2.1e^t$ thousand. What is the instantaneous growth rate at $t = 1.43$?
- (A) 12.549 thousand per unit time (B) 3.003 thousand per unit time
(C) 8.775 thousand (D) 8.775 thousand per unit time
- 26.** A graphing calculator gives $f(1.7) = 2.031$ and $f'(1.7) = -0.846$. Which tangent approximation estimates $f(1.72)$?
- (A) $2.031 - 0.846(1.72)$ (B) $2.031 - 0.846(0.02)$
(C) $2.031 + 1.7(0.02)$ (D) $-0.846 + 2.031(0.02)$
- 27.** Use a calculator to find $f'(1.08)$ for $f(x) = \sin x + e^x$.
- (A) 3.652 (B) 3.827
(C) 2.416 (D) 3.416
- 28.** A calculator gives $f(1.99) = 3.9401$, $f(2) = 4$, and $f(2.01) = 4.0601$. The best symmetric estimate of $f'(2)$ is
- (A) 3 (B) 4
(C) 12 (D) 6
- 29.** A population is $P(t) = 4.6e^t$ thousand. What is the instantaneous growth rate at $t = 1.77$?
- (A) 27.006 thousand per unit time (B) 47.8 thousand per unit time
(C) 8.142 thousand per unit time (D) 27.006 thousand
- 30.** Use a calculator to evaluate the instantaneous rate of change of $f(x) = 1.65x^2 - 1.90x$ at $x = 2.84$.
- (A) 8.472 (B) 7.472
(C) 7.912 (D) 1.4
- 31.** Use a calculator to find $f'(0.96)$ for $f(x) = \sin x + e^x$.
- (A) 3.185 (B) 2.185
(C) 3.431 (D) 3.081
- 32.** A calculator table gives $f(1.6) = 3.7$, $g(1.6) = 1.4$, $f'(1.6) = 0.8$, and $g'(1.6) = 1.4$. Find $(fg)'(1.6)$.
- (A) 5.3 (B) 7.3
(C) 6.3 (D) -6.3
- 33.** A tangent to $y = \ln x$ is drawn at $x = 3.02$. What is its slope to three decimal places?
- (A) 1.105 (B) 0.331
(C) 3.02 (D) -0.331
- 34.** Use a calculator to find $f'(0.67)$ for $f(x) = \sin x + e^x$.
- (A) 2.093 (B) 1.738
(C) 2.738 (D) 2.575
- 35.** A calculator table gives $f(2.4) = 4.4$, $g(2.4) = 2.3$, $f'(2.4) = -2.7$, and $g'(2.4) = -1.8$. Find $(fg)'(2.4)$.
- (A) -13.13 (B) -15.13
(C) 14.13 (D) -14.13
- 36.** Use a calculator to evaluate the instantaneous rate of change of $f(x) = 2.14x^2 + 0.24x$ at $x = 2.18$.
- (A) 9.57 (B) 10.693
(C) 4.52 (D) 10.57

Section II – Free Response

Questions 1–2 may be completed with a calculator. Questions 3–4 are non-calculator.

FRQ1 – Rates, derivative estimates, and tangent model

A differentiable function P models a population in thousands. Selected values are					
t	1.8	1.9	2.0	2.1	2.2
$P(t)$	4.612	4.801	5.000	5.209	5.428

(a) Estimate $P'(2)$ using the best symmetric difference available. Include units. [2 points]

(b) Write the tangent-line model at $t = 2$ and use it to estimate $P(2.06)$. [3 points]

(c) Compare the one-sided secant estimates from 1.9 to 2 and from 2 to 2.1. What does the comparison suggest about $P''(2)$? [2 points]

(d) Explain why the table alone does not prove differentiability at $t = 2$. [1 points]

FRQ2 – Derivative values from a table

At $x = 1$, $f = 3$, $f' = -2$, $f'' = 5$, $g = 4$, $g' = 6$, and $g'' = -1$.

(a) Find $(fg)'(1)$ and $(f/g)'(1)$. [4 points]

(b) Find $(fg)''(1)$. [3 points]

(c) Write the tangent line to $h(x) = f(x)g(x)$ at $x = 1$.

[2 points]

FRQ3 – Derivative from first principles

Let $f(x) = \sqrt{x+5}$.

(a) Use the limit definition, not a memorized rule, to find $f'(a)$ for $a > -5$.

[4 points]

(b) Find the tangent line at $x = 4$.

[2 points]

(c) State the domain of the derivative function and explain the excluded endpoint.

[2 points]

FRQ4 – Continuity and differentiability at a join

Let $p(x) = \begin{cases} ax + b, & x < 1, \\ x^2 - 2x + 4, & x \geq 1. \end{cases}$

(a) Find the relation between a and b required for continuity at 1.

[2 points]

(b) Find a and b so that p is differentiable at 1.

[3 points]

(c) Explain why matching derivatives without matching values is insufficient.

[2 points]

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